

Early- and late-onset preeclampsia as molecularly distinct conditions: a stratified transcriptomic analysis of the placenta and the limits of signature transferability to cfRNA

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Abstract

Background. Preeclampsia is clinically divided into early-onset (<34 weeks) and late-onset (≥ 34 weeks) forms, yet whether these represent molecularly distinct conditions or a severity continuum remains unresolved. This distinction matters for non-invasive testing: whether placental signatures transfer to cell-free RNA (cfRNA) in maternal plasma determines whether predictors can be derived from tissue biology or must be built directly on plasma.

Methods. Were analysed six public NCBI GEO datasets: five placental RNA-seq cohorts (176 samples) for discovery and one cfRNA plasma cohort (199 patients across three sub-cohorts) for validation. Because onset was almost fully confounded with cohort, were used a stratified DESeq2 design rather than batch correction. A compact panel was selected by L1-regularised logistic regression under leave-one-dataset-out cross-validation with in-fold feature selection, benchmarked against random forest and XGBoost, and interpreted with SHAP. The frozen panel was applied to cfRNA; natively trained cfRNA models and differential expression in plasma served as controlled contrasts.

Results. Early- and late-onset preeclampsia shared only four differentially expressed genes and no enriched pathways. The early-onset form showed a hypoxic signature (KEGG HIF-1 signalling, $p\text{-adj} \approx 8 \times 10^{-4}$), independently replicated in a separate early-onset severe cohort with 166 of 166 shared genes concordant in direction. The late-onset form showed membrane-repolarisation and immune processes detectable only in male-bearing placentas (117 versus 15 genes). A 25-gene panel reached a cross-cohort AUC of 0.861 (LASSO), outperforming random forest (0.786) and XGBoost (0.780), with the late-onset cohort scoring lowest in all three models. Transfer to cfRNA was limited (best AUC 0.657, 13–20 weeks), natively trained cfRNA models performed no better (0.613), and none of the 25 panel genes were differentially expressed in plasma.

Availability. All data are publicly available from NCBI GEO; analysis code and accession numbers are available at [repository URL].

1. Introduction

Hypertensive disorders of pregnancy (HDP), and preeclampsia in particular, remain a leading cause of maternal and perinatal morbidity and the principal cause of iatrogenic prematurity. Global prevalence of preeclampsia is approximately 4.4% (95% CI 3.73–5.20), with substantial dependence on the diagnostic criteria applied [1]; HDP account for roughly 18% of maternal deaths worldwide. Incidence is rising: in the United States, HDP affect one in six pregnancies and their frequency nearly doubled between 2007 and 2019 [2]. Because delivery remains the only definitive treatment, clinical value shifts towards early risk prediction.

Preeclampsia is diagnosed from 20 weeks of gestation under ISSHP criteria [3] and is conventionally divided into early-onset (EOPE, <34 weeks) and late-onset (LOPE, ≥ 34 weeks) forms. The classical two-stage model links the early form to inadequate spiral artery remodelling, ischaemia and oxidative stress of the placenta [4, 5]. The revised model extends this framework to the late form, which is frequently preceded by normal placentation but involves placental stress and senescence towards term — arguing that no preeclampsia is purely maternal, since all forms require a placenta [6]. The clinical asymmetry between the forms is nevertheless striking: early preeclampsia has early biophysical markers, is detected well by first-trimester screening and responds to aspirin prophylaxis, whereas the late form runs asymptotically for a long period, is poorly detected and is not prevented by aspirin. The least predictable form is therefore also the least understood molecularly.

Whether EOPE and LOPE are genuinely distinct conditions remains open. Transcriptomic studies of the placenta have identified molecular subclasses of preeclampsia that do not reduce to symptom severity [7]. More recently, a prospective cohort of $n = 9,102$ with blood sampling at 17.5–22 weeks defined two molecular HDP subtypes — placental-associated (led by PAPP2) and immune-associated (led by CD163);

the former corresponds to severe and preterm forms, the latter largely to term preeclampsia and gestational hypertension [8]. Importantly, this axis reclassifies rather than reproduces the clinical early/late division.

Two gaps remain. First, the largest cfRNA studies are conducted on private cohorts that cannot be independently verified. Second, it is unknown whether tissue (placental) signatures transfer into plasma — a question that determines the strategy for developing non-invasive tests. The aim of this work is therefore to use public data to (i) test the molecular distinctness of EOPE and LOPE, (ii) build a compact predictive panel, and (iii) evaluate its transferability to cfRNA.

Hypotheses:

- H1 — EOPE and LOPE have distinct transcriptomic signatures;
- H2 — a compact panel can separate preeclampsia from control in placental tissue across cohorts;
- H3 — the placental panel retains predictive power in plasma cfRNA.

2. Data and methods

Data. Six public NCBI GEO datasets were used (Appendix A, Table A1): five placental RNA-seq cohorts (176 samples) for discovery and one cfRNA plasma cohort (199 patients across three sub-cohorts) for validation. Gene identifiers were mapped to a common gene-symbol space; metadata were unified using an explicit map of diagnosis columns for each dataset.

Branch 1 — stratified differential expression. Because onset is almost fully confounded with cohort in the available data ($\text{EOPE} \approx \text{GSE203507}$, $\text{LOPE} \approx \text{GSE306864}$), merging the matrices with batch correction would have erased the onset-specific signal together with the batch effect. Each dataset was therefore analysed separately (DESeq2, design $\sim$ category; for GSE306864, $\sim$ sex + category), with apeglm shrinkage and thresholds of $p\text{-adj} < 0.05$ and $|\log_2\text{FC}| > 1$. Functional enrichment used clusterProfiler (GO BP and KEGG) with the universe set to all tested genes. Independent replication was performed on an early-onset severe cohort (GSE148241) using the same protocol.

Branch 2 — predictive panel. Normalisation was performed per dataset (log1p followed by z-scoring within each dataset), which makes raw counts, RPKM and normalised values comparable, acts as batch correction, and does not leak across fold boundaries. The feature space was restricted to transferable genes (intersection with the cfRNA cohort, 6,184 genes). Leave-One-Dataset-Out cross-validation was applied, with feature selection (univariate pre-filter plus LASSO) performed inside each fold; LASSO, RandomForest and XGBoost were compared. Interpretation used SHAP (global summary and local waterfall plots).

cfRNA validation. The panel trained on the full placental cohort was frozen and applied to cfRNA; evaluation was at patient level with bootstrap confidence intervals, stratified by sampling timepoint. Separately, models were trained directly on cfRNA (LODO across the three sub-cohorts) in the same feature space, in order to distinguish “the signal does not transfer” from “there is no signal”. Differential expression in plasma was analysed by timepoint (DESeq2, $\sim$ cohort + condition), using one sample per patient in every subset.

Unsupervised check. Consensus NMF (100 subsampling runs) was applied to 35 preeclampsia samples, computing the adjusted Rand index separately against onset and against dataset in order to detect confounding.

3. Results

3.1. EOPE and LOPE are molecularly distinct

Stratified analysis revealed minimal overlap between the subtypes: EOPE (283 DEGs) and LOPE (117 DEGs in male-bearing placentas) share only four genes — all concordant in direction — and no enriched pathway (Table 1).

Table 1. Stratified differential expression analysis

Stratum	n (Case/Control)	DEGs
EOPE (GSE203507)	15 / 5	283
LOPE, both sexes (GSE306864)	13 / 45	15
LOPE, male-bearing only	5 / 16	117
mixed (GSE190971)	7 / 6	452
EOPE $\cap$ LOPE / shared pathways	—	4 (4/4 concordant) / 0

Thresholds: $p\text{-adj} < 0.05$ and $|\log_2FC| > 1$. The contrast between the two LOPE rows shows how pooling fetal sexes masks the signal.

EOPE is characterised by a hypoxic signature: the only significant KEGG pathway is HIF-1 signalling ($p\text{-adj} \approx 8 \times 10^{-4}$, 11 genes), and the leading GO terms are response to hypoxia and response to decreased oxygen levels (Appendix B, Figures B1–B2). LOPE is enriched for membrane-repolarisation and vascular-remodelling processes with no hypoxic term at all (Figure B3). The fetal-sex effect is pronounced: pooling both sexes yielded 15 DEGs versus 117 in the male-only subset, despite three times fewer cases.

3.2. Independent replication of the hypoxic EOPE signature

In an independent early-onset severe cohort (GSE148241), 451 DEGs were identified, of which 166 overlap with the EOPE signature — and all 166 are concordant in direction (100%, against $\approx 50\%$ expected by chance). Hypoxic GO terms replicated independently.

3.3. Predictive panel and model comparison

Cross-cohort performance is shown in Table 2: LASSO achieved a mean AUC of 0.861, RandomForest 0.786 and XGBoost 0.780. The advantage of the linear model is expected under $p \gg n$. Notably, the late-onset cohort produced the lowest AUC in all three models (0.679 / 0.675 / 0.643): a model trained largely on early preeclampsia systematically fails to transfer to the late form.

Table 2. Leave-One-Dataset-Out cross-validation (ROC-AUC)

Held-out dataset	LASSO	RF	XGBoost
GSE203507 (EOPE)	1.000	0.947	1.000
GSE190971 (mixed)	1.000	0.881	0.952
GSE186257 (severe)	0.842	0.818	0.780
GSE148241 (EOSPE)	0.783	0.609	0.526
GSE306864 (LOPE)	0.679	0.675	0.643
Mean AUC	0.861	0.786	0.780

Feature selection was performed inside each fold. Values of 1.000 were obtained on small held-out sets ($n = 13\text{--}20$) and require cautious interpretation.

Two panels were derived: top-25 and stable-18, sharing 14 genes. The leading SHAP features are HK2 (a canonical HIF-1 target, hypoxic axis) and CX3CR1 (fractalkine receptor, immune axis), so the panel spans both subtype axes (Figure B4). Local interpretation (Figure B5) shows two cases with almost identical model output ($f(x) = 0.858$ and 0.854) but different decision structure: one dominated by HK2, the other built from a distributed non-hypoxic profile.

3.4. Limited transferability to cfRNA

The frozen panel performed best in the 13–20-week window (AUC 0.657; 95% CI 0.532–0.764), whereas at the earlier and later timepoints performance was at chance level (Table 3). Models trained natively on cfRNA in the same feature space did not outperform the transfer on early plasma (0.613), indicating a weakness of the plasma signal itself rather than a problem of transfer alone.

Table 3. cfRNA validation: transfer versus natively trained models (ROC-AUC [95% CI])

Timepoint	n (PE)	Transfer	Native (6,184)	Native (7,109)
≤12 weeks	127 (32)	0.545	—	—
13–20 weeks	113 (34)	0.657 [0.53–0.76]	0.613 [0.52–0.70]	0.598 [0.50–0.69]
≥23 weeks	82 (26)	0.544	0.648 [0.52–0.77]	0.707 [0.59–0.81]

Patient-level evaluation, one sample per patient; bootstrap CIs by resampling patients.

Differential expression in plasma confirmed the gap at gene level: of 93 cfRNA DEGs identified (0 at ≤12 weeks, 76 at 13–20 weeks, 26 at ≥23 weeks), only 2 of 283 EOPE DEGs, 3 of 117 LOPE DEGs and none of the 25 panel genes overlap with the placental signatures. Pooling timepoints sharply reduced the number of DEGs (4 across 196 samples), indicating that the plasma signature is timepoint-specific. Functional enrichment of the 13–20-week set returned no significant terms, and the dominance of XIST and myeloid markers (ITGAX, MPEG1, CD300E) points to differences in cell composition and fetal fraction.

3.5. Unsupervised clustering does not provide independent confirmation

Consensus NMF produced highly stable clusters (cophenetic > 0.94) that align perfectly with onset (ARI up to 1.0). However, the ARI against onset is identical to the ARI against dataset for every k, and the mixed samples form a separate cohort-specific cluster (Table A2, Figure B6). Because of this confounding, the analysis cannot serve as independent confirmation of the subtypes.

4. Discussion

Hypothesis H1 is supported at several independent levels: the separation of subtypes is reproduced at the level of genes (an overlap of four DEGs), mechanisms (no shared pathways) and predictive behaviour of the models. The strongest argument is the replication of the hypoxic EOPE signature in an independent cohort with perfect directional concordance, which excludes dataset-specific noise.

Mechanistically, the hypoxic signature of EOPE is consistent with the classical two-stage model of pathogenesis [4, 5], and the presence of HK2 among the leading panel features supports this interpretation by an independent method. LOPE, by contrast, has no hypoxic component and is characterised by membrane and immune processes — consistent with the revised model, in which late-onset disease arises from placental stress at term rather than from ischaemic origin [6]. The axis obtained here — hypoxic-placental EOPE versus immune-membrane LOPE — is consistent with the subtypes defined in a large prospective cfRNA cohort, where more severe forms were associated with placental genes (PAPPA2) and less severe forms with immune genes (CD163) [8]. Importantly, in that study the molecular axis reclassifies the clinical early/late division: early-onset cases separate depending on whether they lead to preterm delivery. Our results, obtained in tissue and on independent open data, reproduce the same biological dichotomy while remaining anchored to clinical onset, which is consistent with correlation rather than identity between the two axes.

The fetal-sex specificity of LOPE is consistent with evidence on placental sexual dimorphism: in severe preeclampsia, cytokine production, apoptosis and NFκB p65 activity are increased in male-bearing placentas relative to female-bearing ones [9]. The primary study of the cohort used here independently reported that differential expression is detectable only in male-bearing placentas, with deconvolution indicating increased proportions of CD14⁺ monocytes and activated CD8⁺ T cells [10]. Proposed mechanisms include dosage effects of genes escaping X-inactivation, a more aggressive growth trajectory and lower functional reserve of the male placenta with vulnerability to lipotoxicity, and sex differences in the phenotype and metabolism of placental macrophages [11, 12].

Hypothesis H3 was not supported, and the manner of its refutation is itself informative. The controlled contrast showed that models trained natively on cfRNA do not outperform the transfer on early plasma, indicating that the limitation lies less in moving tissue signatures than in the weakness of the early plasma signal in these data. Differential expression confirmed the compartment gap at gene level (0 of 25 panel genes among plasma DEGs). At the same time, the dominance of XIST and myeloid markers in the cfRNA

signature indicates an influence of cell composition — a limitation that clinically validated studies address by explicit correction for fetal fraction and fetal sex [8]; the absence of such correction here is a direct specification for future work.

This work does not propose a diagnostic tool and does not compete with existing solutions: current first-trimester screening achieves high detection of early preeclampsia, and a cfRNA classifier has already undergone prospective validation (AUC 0.74 in the population without high risk factors; 0.88 in the advanced-maternal-age subgroup, $n = 403$ with 11 cases; NPV 99.7%, PPV 8.9%) [8]. The contribution lies in an independent, reproducible confirmation of the subtyping hypothesis on open data, and in the empirical argument for building non-invasive predictors directly on plasma.

The principal direction arising from this work is the prioritisation of late-onset preeclampsia. Our data add a molecular dimension to the clinical asymmetry: LOPE was the hardest form to detect in every model, has its own non-hypoxic biology, and manifests in a fetal-sex-specific manner, while the available tissue reflects the state after clinical manifestation. The limitations of this study with respect to LOPE are therefore better read as a specification for future designs: longitudinal cfRNA cohorts with early serial sampling, mandatory fetal-sex stratification, cell-composition deconvolution, and larger samples. This does not imply that early preeclampsia is exhausted as a subject — it remains the leading cause of iatrogenic prematurity; the point is a rebalancing of attention proportional to clinical burden.

5. Limitations

- Confounding of onset with cohort is the principal design limitation; it precludes matrix merging and independent verification of the subtypes by clustering.
- Small sample sizes: results for EOPE are robust, whereas those for LOPE (male-only, $n = 5$ cases) and for the mixed cohort are preliminary; AUC values of 1.000 on held-out sets of $n = 13$ –20 should be interpreted with caution.
- The temporal gap between tissue (collected at delivery) and cfRNA (early plasma) constrains translational conclusions.
- Heterogeneous normalisation across the ML pool, partially mitigated by per-dataset z-scoring.
- cfRNA analysis was performed without cell-composition deconvolution or fetal-fraction correction.
- cfRNA validation rests on a single cohort (three sub-cohorts). A second cfRNA cohort (GSE154377) and a microarray cohort (GSE75010) were additionally downloaded but not integrated into the analyses; one further dataset (GSE97320) was excluded as irrelevant (peripheral blood microarray, myocardial infarction, $n = 6$).

6. Conclusions

1. EOPE and LOPE are molecularly distinct conditions: they share only four differentially expressed genes and no enriched pathway, and the distinction is confirmed at the level of genes, mechanisms and predictive behaviour of models.
2. The hypoxic signature of early preeclampsia (HIF-1 signalling) is reproducible: replication in an independent cohort yielded 166 shared genes with 100% directional concordance.
3. Late preeclampsia has its own non-hypoxic — membrane and immune — biology and manifests exclusively in male-bearing placentas.
4. A 25-gene panel separates preeclampsia in placental tissue across cohorts (AUC 0.861) but transfers poorly to cfRNA; the compartment gap is confirmed at gene level (0 of 25).
5. Non-invasive predictors of preeclampsia should be built directly on cfRNA, with correction for fetal fraction and fetal sex.

6. Late-onset preeclampsia — the form hardest to detect molecularly and least served clinically — requires prioritised, fetal-sex-stratified study in longitudinal cfRNA cohorts.

Appendix A. Supplementary tables

Table A1. Cohort characteristics

GSE	Technology	n	Format	Role	Onset
GSE203507	RNA-seq	20*	raw counts	DEA + ML	EOPE
GSE306864	RNA-seq	58	raw counts	DEA + ML	LOPE
GSE190971	RNA-seq	13	raw counts	DEA + ML	mixed
GSE186257	RNA-seq	44	DESeq2-norm	ML	severe
GSE148241	RNA-seq	41	raw counts	ML + replication	EOSPE
GSE192902	RNA-seq	199†	raw counts	cfRNA validation	—

* villous-only after tissue filtering (from 57). † unique patients after deduplication of longitudinal draws; the cohort comprises three sub-cohorts used as folds in the native cfRNA analysis. The table lists only datasets used in the analyses.

Table A2. Consensus NMF: clusters versus onset and cohort

k	Cophenetic	Silhouette	ARI (onset)	ARI (dataset)
2	0.940	0.314	0.693	0.693
3	0.988	0.436	1.000	1.000
4	0.993	0.412	0.916	0.916

The identity of ARI against onset and against dataset makes it impossible to distinguish a biological split from a cohort effect.

Table A3. cfRNA differential expression and overlap with placental signatures

Subset / overlap	n (Control/PE)	Value
≤12 weeks	95 / 32	0 DEGs
13–20 weeks	79 / 34	76 DEGs
≥23 weeks	56 / 26	26 DEGs
pooled antenatal	139 / 57	4 DEGs
Union of subsets	—	93 DEGs
$\cap$ EOPE DEGs / LOPE DEGs	—	2 / 283 · 3 / 117
$\cap$ panel genes (top-25)	—	0 / 25

DESeq2, design ~ cohort + condition, one sample per patient.

Appendix B. Figures

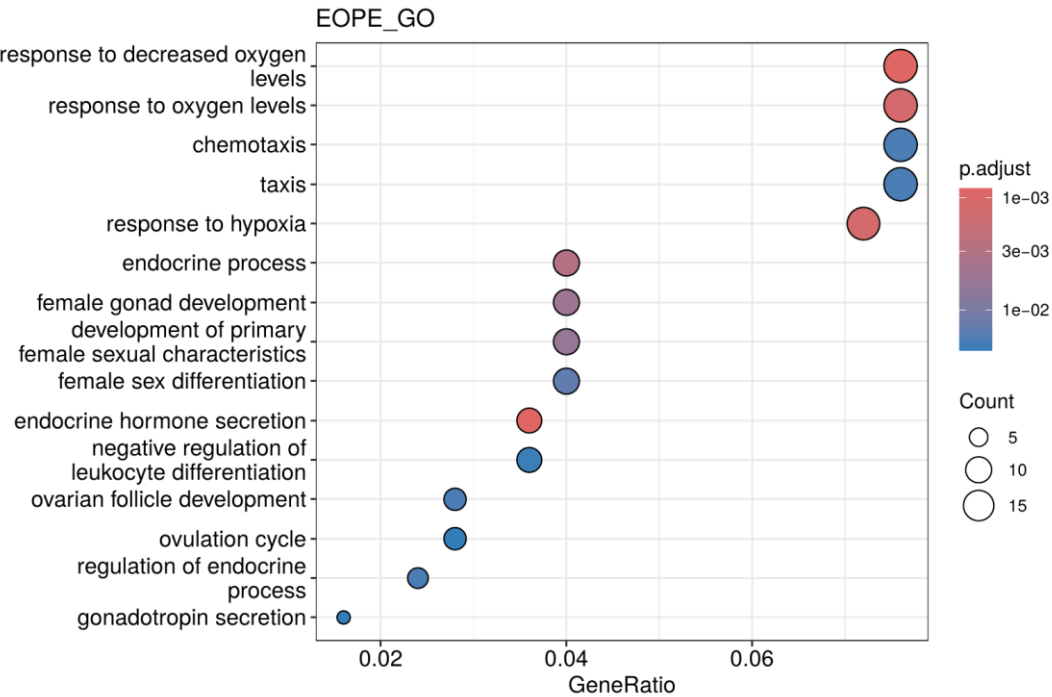


Figure B1. GO (biological process) enrichment for the 283 differentially expressed genes of early-onset preeclampsia. Terms relating to hypoxia and decreased oxygen levels dominate the ranking. Dot size reflects the number of genes; colour reflects the adjusted p-value. Universe: all tested genes.

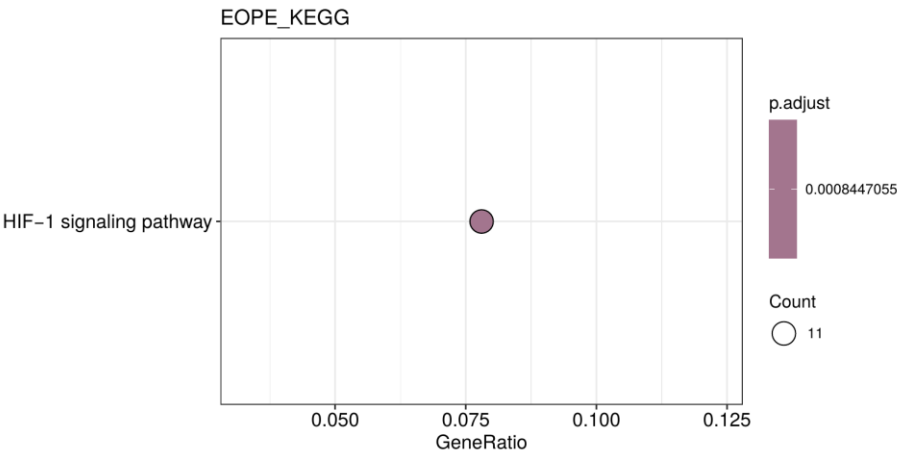


Figure B2. KEGG enrichment for early-onset preeclampsia. HIF-1 signalling is the only significant pathway ($p\text{-adj} \approx 8 \times 10^{-4}$, 11 genes), consistent with the two-stage model of placental ischaemia.

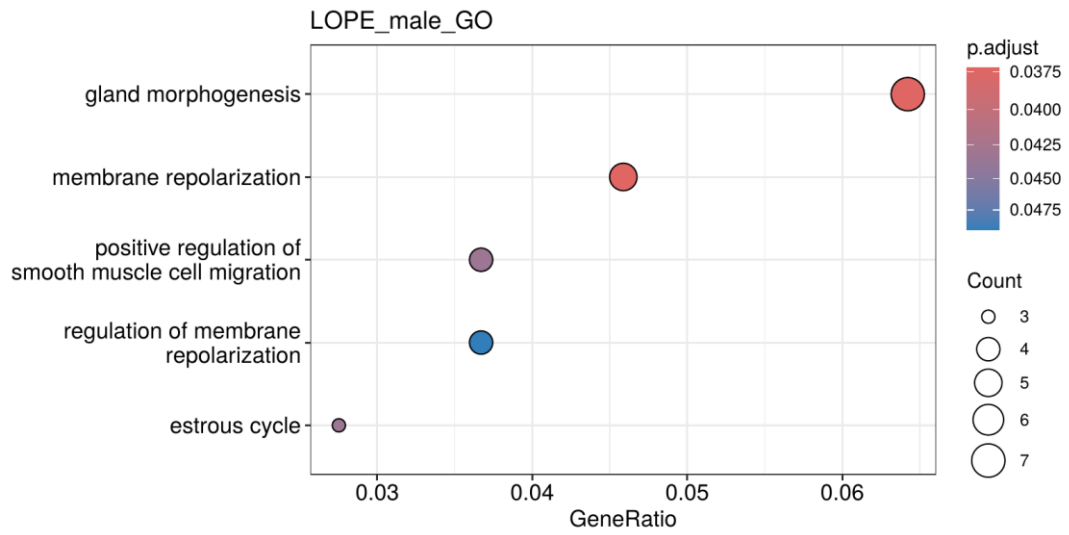


Figure B3. GO (biological process) enrichment for the 117 differentially expressed genes of late-onset preeclampsia in male-bearing placentas. Membrane repolarisation and vascular remodelling processes are recovered; no hypoxic term appears, and no term is shared with Figure B1. Adjusted *p*-values (0.037–0.048) lie close to the significance threshold, reflecting the limited sample size.

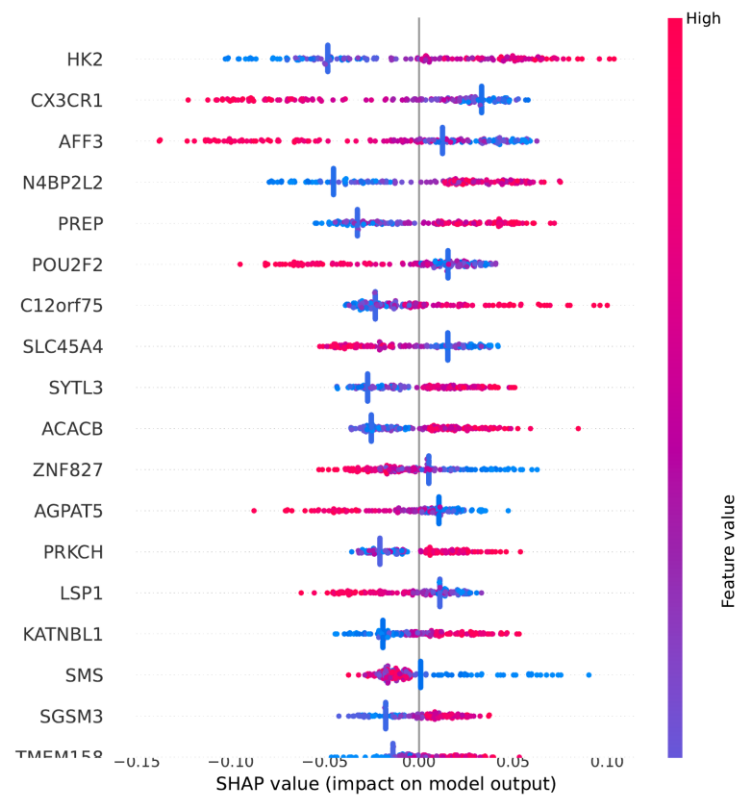


Figure B4. Global SHAP feature importance for the 25-gene panel (RandomForest, TreeExplainer; top 16 features shown). The leading features are HK2 (hexokinase 2, a canonical HIF-1 target — the hypoxic axis) and CX3CR1 (fractalkine receptor — the immune axis), indicating that the panel spans both subtype axes. Colour encodes the expression value; position on the x-axis encodes the direction and magnitude of the contribution to the prediction.

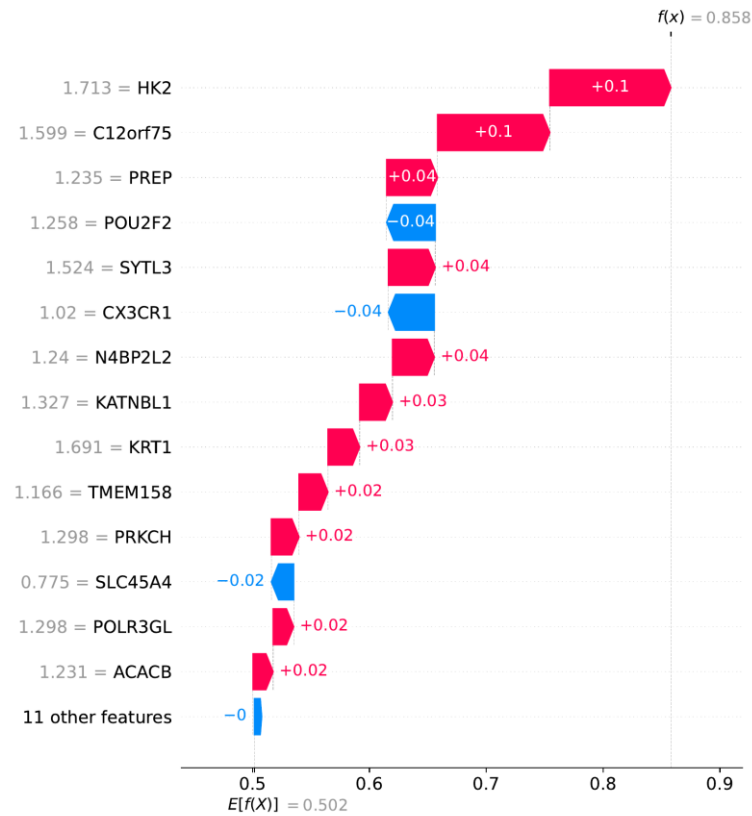


Figure B5a. Local SHAP interpretation, hypoxic (EOPE-like) profile. Model output $f(x) = 0.858$, driven primarily by HK2 (+0.10).

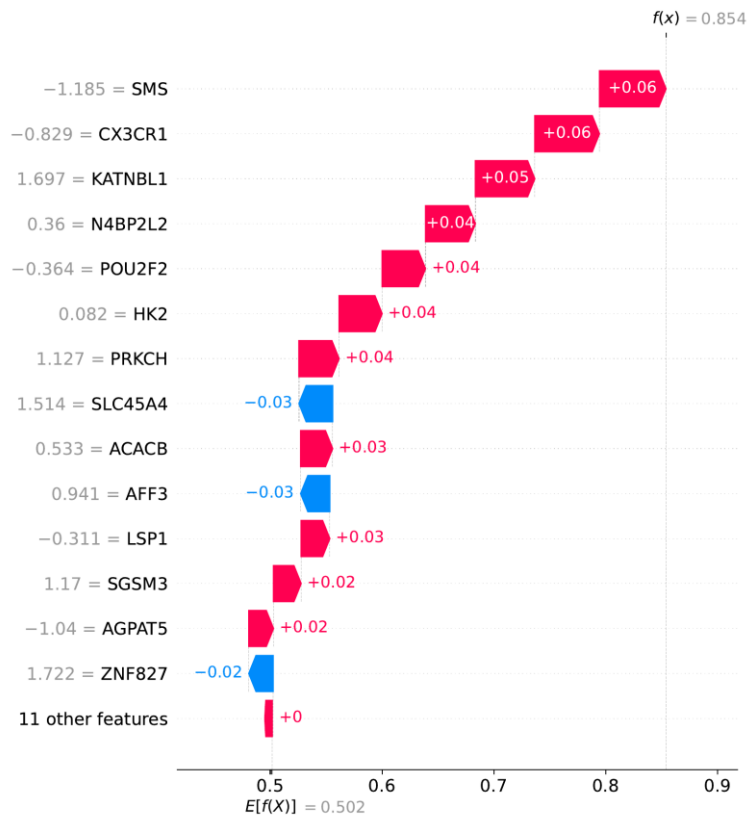


Figure B5b. Local SHAP interpretation, immune (LOPE-like) profile. Model output $f(x) = 0.854$ — practically identical to B5a — but the contribution of HK2 is minimal (+0.03) and the prediction is built from a distributed non-hypoxic profile. Together, B5a and B5b illustrate molecular heterogeneity at the level of an individual patient.

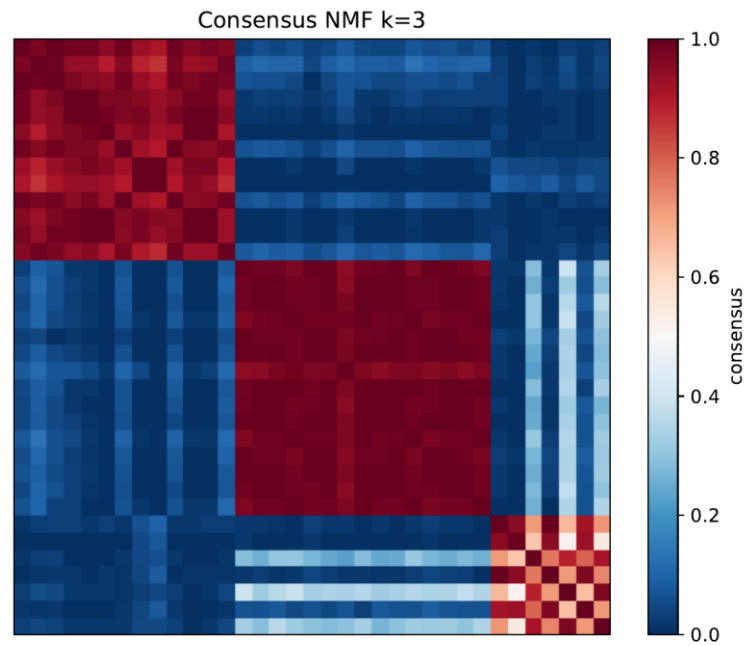


Figure B6. Consensus matrix of unsupervised NMF clustering of preeclampsia samples (100 subsampling runs, $k = 3$). Despite high cluster stability ($cophenetic > 0.94$), the structure coincides identically with both onset and cohort, so the analysis cannot serve as independent confirmation of the subtypes.

Appendix C. Reproducibility

- The pipeline is organised into functional stages, from data acquisition through to validation; scripts and dependency files are provided in the project repository.
- Python: GEParse, pandas, scikit-learn, shap, mygene, xgboost. R: DESeq2, apegglm, clusterProfiler, org.Hs.eg.db.
- Full DEG lists and the composition of the top-25 and stable-18 panels are provided in the results directory.

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